

E.G.S.PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai
Accredited by NAA C with A++ Grade | Accredited by NBA(CSE, ECE, IT)

NAGAPATTINAM-611002



B.E. Biomedical Engineering

First Year – Second Semester

Course Code	Course Name	L	T	P	C	Maximum Marks		
						CA	ES	Total
Theory Course								
1901MA204	Engineering Mathematics–II (Calculus, Ordinary Differential Equations and Complex Variable)	3	2	0	4	40	60	100
1901PH202	Semiconductor Physics and Optoelectronics	3	0	0	3	40	60	100
1901GEX01	Basic Electrical and Electronics Engineering	3	0	0	3	40	60	100
1901GEX02	Engineering Graphics	2	2	0	3	50	50	100
1901GE201	Engineering Exploration	2	0	0	2	40	60	100
Laboratory Course								
1901GE254	Computer Hardware and IT Essentials Lab	0	0	2	1	50	50	100
1901GE252	Engineering Intelligence -II	0	0	2	1	100	0	100
1901GEX51	CAD Lab	0	0	2	1	50	50	100
1901GEX53	Basic Electrical and Electronics Engineering Lab	0	0	2	1	50	50	100
1901PHX51	Engineering Physics Lab	0	0	2	1	50	50	100

L–Lecture |T–Tutorial |P–Practical |CA–Continuous Assessment |ES–End Semester

Module IV	COMPLEX VARIABLE–DIFFERENTIATION	12 Hours
Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; Conformal mappings, Mobius transformations.		
Module V	COMPLEX VARIABLE– INTEGRATION	12 Hours
Contour integrals, Cauchy Integral formula (without proof), Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using The Bromwich contour.		
Total:60Hours		
TEXTBOOKS:		
1. G.B.Thomas and R.L.Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002. 2. Erwin kreys zig, Advanced Engineering Mathematics, 9th Edition, John Wiley& Sons, 2006. 3. W.E.Boyce and R.C.Di Prima, Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009. 4. S.L.Ross, Differential Equations, 3rd Ed., Wiley India, 1984. 5. J.W.Brown and R.V.Churchill, Complex Variables and Applications, 7th Ed., McGrawHill, 2004. 6. N.P.Bali and Manish Goyal, A textbook of Engineering Mathematics, Laxmi Publications, Reprint, 2008. 7. B.S.Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.		

1901PH202	SEMICONDUCTOR PHYSICS AND OPTO ELECTRONICS (for ECE and BME)										L	T	P	C
											3	0	0	3
COURSE OBJECTIVES:														
1.To instill knowledge on physics of semiconductors, determination of charge carriers and device applications														
2.To make the students to understand the basics of dielectric materials, electrical properties of materials including free electron theory, applications of quantum mechanics														
3.To establish a sound grasp of knowledge on different magnetic & optical properties of materials														
4.To make the students to understand the basics of optoelectronic devices, optical displays and applications														
5.To inculcate an idea of significance of nano structures, quantum confinement and ensuing nano device applications.														
PREREQUISITES: Basic knowledge in physics														
COURSE OUTCOMES;														
Upon completion of this course, students will be able to														
CO1: Apply the conditions of energy states of electrons and energy band of materials and its Fermi level														
CO2: Determinethetypeofsemiconductingmaterial,itsenergygapandcarrierconcentration														
CO3: Apply the conditions for semiconductor lasers in GaAs and other tunable lasers														
CO4: Experiment with PN junctions and its applications in solar cells CO5:Apply quantum confinement concepts to QD based LEDs and Lasers														
COs Vs POs MAPPING:2														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		
CO1	3	2	2	1	-	-	-	2	2	-	-	1		
CO2	3	2	2	1	-	-	-	2	2	-	-	1		
CO3	3	2	2	1	-	-	-	2	2	-	-	1		
CO4	3	2	2	1	-	-	-	2	2	-	-	1		
CO5	3	2	2	1	-	-	-	2	2	-	-	1		
AVG	3	2	2	1	-	-	-	2	2	-	-	1		
COs Vs PSOs MAPPING:														
COs	PSO1	PSO2	PSO3											
CO1	1	-	-											
CO2	1	-	-											
CO3	1	-	-											
CO4	1	-	-											
CO5	1	-	-											
AVG	1	-	-											
Module I	ELECTRONIC MATERIALS											9 Hours		
Free electron theory, Density of states and energy band diagrams, Energy bands in solids, E-k diagram,Directandindirectbandgaps,Typesofelectronicmaterials:metals,semiconductorsand insulators, Occupation probability, Fermi level0														
Module II	SEMICONDUCTORS											9 Hours		
Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier- concentration and temperature (equilibrium carrier statistics), Carrier generation and recombination, Carrier transport: diffusion and drift, p-n junction, Metal- semiconductor junction(Ohmic and Schottky)														
Module III	SEMICONDUCTOR LASERS											9 Hours		
Optical transitions in bulk semiconductors: absorption, spontaneous emission, and stimulated emission; Density of states for photons, Transition rates (Fermi's golden rule), Optical loss and gain. Semiconductor														

laser(GaAs):materials, device characteristics, figures of merit and Vertical- Cavity Surface-Emitting Lasers (VECSEL), Tunable semiconductor lasers		
Module IV	SEMICONDUCTOR PHOTO DETECTORS	9 Hours
Types of semiconductor photo detectors-p-n junction, PIN ,and Avalanche and their structure, working principle, and characteristics, Noise limits on performance; Solar cells.		
Module V	NANO-OPTO ELECTRONIC DEVICES	9 Hours
Quantum well, quantum wire, and- dot based LEDs, white light LED lasers, and photo detectors.		
References:		
1. J.Singh, Semiconductor Opto electronics: Physics and Technology, McGraw-Hill Inc.(1995). 2. B.E.A.SalehandM.C.Teich,FundamentalsofPhotonics,JohnWiley&Sons,Inc.,(2007). 3. S.M.Sze, Semiconductor Devices: Physics and Technology, Wiley (2008). 4. A.Yariv and P. Yeh, Photonics: Optical Electronics in Modern Communications, Oxford University Press, New York (2007). 5. P.Bhattacharya, Semiconduct or Opto electronic Devices, Prentice Hall of India(1997). 6. Online course:"Semiconductor Optoelectronics "by MR Shenoyon NPTEL 7. Online course:"Opto electronic Materials and Devices" by Monica Katiyar and Deepak Guptaon NPTEL		
References(WEBSITES):		
1. https://www.daenotes.com/electronics/microwave-radar/semiconductor-laser 2. https://iopscience.iop.org/article/10.1088/0957-0233/12/5/703 3. https://www.elprocus.com/photodiode-working-principle-applications/ 4. https://www.azoquantum.com/Article.aspx?ArticleID=31 5. https://www.understandingnano.com/quantum-dots-applications.html		

1901GEX01	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING (Common for all UG programmes , except BE-EEE)	L	T	P	C
		3	0	0	3

COURSE OBJECTIVES::

- 1.To introduce basic DC and AC circuits
2. To impart knowledge in the basic working principles and applications of electrical machines and measuring instruments
3. To educate the fundamental concepts of analog and digital electronics.

PREREQUISITES: Basic knowledge in electrical and electronics

COURSE OUTCOMES

Upon completion of this course ,students will be able to

1. Solve very simple problems in DC and AC circuits
2. Explain the construction and principle of operation of DC and AC machines
3. Describe the operation of simple electrical measuring instruments
4. Elucidate the characteristics of diode, Zener diode, BJT, SCR and their applications
5. Implement Boolean expressions using logic gates and Explain the operation of functional blocks of various communication systems

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	2	-	-	-	-	-	-	2
CO2	3	3	3	-	2	-	-	-	-	-	-	2
CO3	3	3	3	-	2	-	-	-	-	-	-	2
CO4	3	3	3	-	2	-	-	-	-	-	-	2
CO5	3	3	3	-	2	-	-	-	-	-	-	2
AVG	3	3	3	-	2	-	-	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	-	-
CO2	1	-	-
CO3	1	-	-
CO4	1	-	-
CO5	1	-	-
AVG	1	-	-

COURSECONTENTS

Module I	INTRODUCTION TO DC AND AC CIRCUITS	9 Hours
Ohms law - Kirchhoff's laws - Mesh analysis - Nodal analysis - Generation of AC waveforms - Analysis of R-L, R-C, R-L-C circuits - Introduction to three phase systems - Types of connections. Electrical Machines: DC Generator, DC Motor, Transformer, Induction Motor: Working principle, construction and applications.		
Module II	MEASURING INSTRUMENTS	9 Hours
Classification of instruments; Voltmeter, Ammeter, Wattmeter, Energy meter, Multimeter, CRO: Principles and operation. Semiconductor devices: V-I characteristics of PN junction diode and Zener diode; Rectifiers - Half wave and full wave rectifiers; BJT -configurations; Amplifiers & Oscillators: classification, operation and applications; SCR: Construction and V-I characteristics; Basic power converters (Block diagram approach only).		
Module III	DIGITAL SYSTEMS	9 Hours
Boolean algebra - Reduction of Boolean expressions - De-Morgan's theorem - Logic gates - Implementation of Boolean expressions.		
Module IV	COMMUNICATION SYSTEMS	9 Hours
Model of communication system - Analog and digital, Wired and wireless channel - Block diagram of various		

communication systems - Microwave, satellite, optical fiber and cellular mobile system.		
Module V	ELECTRICAL SAFETY AND WIRING	9 Hours
Safety measures in electrical system - Safety devices - types of wiring - Wiring accessories- staircase, fluorescent lamps and corridor wiring- Basic principles of earthing -Types of earthing -layout of generation, Transmission and distribution of power(Single line diagram).		
		Total 45 Hours
References(BOOKS):		
1. Smarajit Ghosh, “Fundamentals of Electrical and Electronics Engineering”, 2nd Edition, PHI Learning, 2010. 2. R.Muthu subramaniam, S.Salaivahanan and K.A.Mureleedharan, “Basic Electrical Electronics and Computer Engineering”, Tata McGraw Hill, 2004. 3. D.P.Kothari and I.J.Nagrath, “Theory and Problems of Basic Electrical Engineering”, PHI learning, New Delhi, 2004. 4. J.B.Gupta, “Fundamentals of Electrical Engineering and Electronics”, S.K.Kataria and Sons, Reprint 2012 Edition. 5. R.L.Boylestad and L.Nashelsky, “Electronic Devices and Circuit Theory”, Pearson, 11th Edition, 2013. 6. George Kennedy and Bernard Davis, “Kennedy’s Electronic communication Systems”, McGraw Hill Education, 5th Edition, 2011. 7. Donald P. Leach, Albert Paul Malvino and Goutam Saha, “Digital Principles and Applications”, McGraw-Hill Education, 8th Edition, 2014.		
References(WEBSITES):		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/downloads/108105053/ 3. https://nptel.ac.in/courses/117103063/ 4. https://nptel.ac.in/courses/117102059/		

1901GEX02	ENGINEERING GRAPHICS								L	T	P	C
									2	0	2	3
COURSEOBJECTIVES:												
1. To develop students' graphic skills for effective communication of concepts, ideas, and designs of engineering products.												
2. To expose students to existing national standards related to technical drawings.												
Prerequisite:												
1. Basic knowledge about geometry												
2. Lettering and Dimensioning												
COURSEOUTCOMES:												
On successful completion of the course, students will be able to:												
CO1: Perform freehand sketching of basic geometrical constructions and multiple views of objects.												
CO2: Do orthographic projections of lines and plane surfaces.												
CO3: Draw projections of solids and develop surfaces.												
CO4: Prepare isometric and perspective sections of simple solids.												
CO5: Demonstrate skills in computer-aided drafting (CAD).												
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	-	-	2	2	-	-	-
CO2	3	3	2	2	1	-	-	2	2	-	-	-
CO3	3	3	2	2	1	-	-	2	2	-	-	-
CO4	3	3	2	2	1	-	-	2	2	-	-	-
CO5	3	3	2	2	1	-	-	2	2	-	-	-
AVG	3	3	2	2	1	-	-	2	2	-	-	
COs Vs PSOs MAPPING:												
</												

Basic Geometrical constructions, Curves used in engineering practices: Conics – Construction of ellipse, parabola and hyperbola by eccentricity method – Construction of cycloid – construction of involutes of square and circle – Drawing of tangents and normal to the above curves. Visualization concepts and Free Hand sketching: Visualization principles –Representation of Three-Dimensional objects–Layout of views–Free hand sketching of multiple views from pictorial views of Objects.			
MODULE III	PROJECTION OF POINTS, LINES AND PLANES SURFACES		9 Hours
Orthographic projection-principles-Principal Planes-First angle projection-projection of points. Projection of straight lines (only First angle projections) inclined to both the principal planes - Determination of true lengths and true inclinations by rotating line method and traces. Projection of planes (polygonal and circular surfaces) inclined to both the principal planes by rotating object method.			
MODULE IV	PROJECTION OF SOLIDS		9 Hours
Projection of simple solids such as prisms, pyramids, cylinders, and cones when the axis is inclined to one of the principal planes, using the rotating object method.			
MODULE V	PROJECTION OF SECTIONED SOLIDS AND DEVELOPMENT OF SURFACES		9 Hours
Sectioning of above solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section. Development of lateral surfaces of simple and sectioned solids – Prisms, pyramids cylinders and cones.			
MODULE VI	ISOMETRIC AND PERSPECTIVE PROJECTIONS		9 Hours
Principles of isometric projection – isometric scale – Isometric projections of simple solids and truncated solids - Prisms, pyramids, cylinders, cones- combination of two solid objects in simple vertical positions and miscellaneous problems. Perspective projection of simple solids-Prisms, pyramids and cylinders by visual ray method.			
TOTAL: 45+5 HOURS			
REFERENCES:			
1. Gopala Krishna K. R., Engineering Drawing (Vol. I & II combined), Subhas Stores, Bangalore, 2016. 2. Luzzader, Warren J., and Duff, John M., Fundamentals of Engineering Drawing with an Introduction to Interactive Computer Graphics for Design and Production, Eastern Economy Edition, Prentice Hall of India Pvt. Ltd, New Delhi, 2005. 3. Shah M. B., and Rana B. C., Engineering Drawing, Pearson, 2nd Edition, 2015. 4. Venugopal K., and Prabhu Raja V., Engineering Graphics, New Age International (P) Limited, 2017. 5. Natrajan K. V., A Textbook of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2015. 6. Basant Agarwal and Agarwal C. M., Engineering Drawing, Tata McGraw Hill Publishing Company Limited, New Delhi, 2008. 7. Bhatt N. D., and Panchal V. M., Engineering Drawing, Charotar Publishing House, 50th Edition, 2016.			

		L	T	P	C
1901GE201	ENGINEERING EXPLORATION	2	0	0	2

Prerequisite: 1. Basic knowledge about Engineering products
2. Willingness to design and fabricate simple products

COURSE OBJECTIVES:

1. Build mindsets and foundations essential for designers.
2. Learn about the Human-Centered Design methodology and understand its real-world applications.
3. Use Design Thinking as a problem-solving methodology to investigate ill-defined problems.
4. Undergo several design challenges and work towards the final design challenge.
5. Apply Design Thinking in the following project streams:
 - **Project Stream 1:** Electronics, Robotics, IoT, and Sensors
 - **Project Stream 2:** Computer Science and IT Applications
 - **Project Stream 3:** Mechanical and Electrical Tools
 - **Project Stream 4:** Eco-friendly solutions for waste management, infrastructure, safety, alternative energy sources, agriculture, environmental science, and other engineering fields.

COURSE OUTCOMES:

CO1: Describe class objectives & harness the designer mindset

CO2: Describe the design thinking methodology

CO3: Demonstrate about how to brainstorm effectively

CO4: Design and fabricate effective prototypes

CO5: Inspect prototypes and ideas through user feedback

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	3	3			-	-	2
CO2	3	3	2	-	-	3	3			-	-	2
CO3	3	3	2	-	-	3	3			-	-	2
CO4	3	3	2	-	-	3	3			-	-	2
CO5	3	3	2	-	-	3	3			-	-	2
AVG	3	3	2	-	-	3	3			-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	2	2	-
CO2	2	2	-
CO3	2	2	-
CO4	2	2	-
CO5	2	2	-
AVG	2	2	-

How to Pursue the Project Work

- a. The first part will be learning-based, guiding students to embrace the methodology by exploring all the phases of Design Thinking through activities like the wallet/bag challenge and podcasts.
- b. The second part will be discussion-based, focusing on building essential skills for designers and learning about complementary materials related to Human-Centered Design.
- c. The class will then be divided into teams, which will collaborate for approximately 2–3 weeks. These teams and design challenges will form the basis for the final project and presentation.
- d. Teams will start with a Design Challenge and go through all phases in depth—from framing the right question, empathizing, ideating, prototyping, to testing.
- e. Outside of class, students will gather requirements, identify challenges, assess usability, and understand the importance of their design solutions.
- f. At the end, students are required to submit the final reports, which will be evaluated by the faculty.

Tasks to Be Done

Task 1: Everyone is a Designer

- Understand class objectives and harness the designer mindset.

Task 2: The Wallet/Bag Challenge and Podcast

- Gain a quick introduction to the Design Thinking methodology.
- Go through all stages of the methodology through a simple design challenge.
- **Podcast:** Observe, listen, and engage with the surrounding environment to identify a design challenge.

Task 3: Teams & Problems

- Start the Design Challenge and learn about teams and problem identification through this process.
- Foster team collaboration, find inspiration from the environment, and learn how to identify problems.

Task 4: Empathizing

- Continue the Design Challenge and focus on learning empathy.
- Learn techniques on how to empathize with users.
- Go to the field and interview people in their environments.
- Submit Activity Card.

Task 5: Ideating

- Continue the Design Challenge and learn how to brainstorm effectively.
- Encourage exploration and foster spaces for brainstorming.
- Submit Activity Card.

Task 6: Prototyping

- Continue the Design Challenge and learn how to create effective prototypes.
- Build tangible models and use them as communication tools.
- Start giving constructive feedback to classmates and teammates.
- Submit Activity Card.

Task 7: Testing

- Finish the Design Challenge and iterate prototypes and ideas through user feedback.
- Evolve ideas and prototypes based on user feedback and constructive criticism.
- Get peer feedback on individual and group performance.
- Submit Activity Card.

Task 8: Final Report Submission and Presentation

- Submit the final report and present the project.
- **Method of Evaluation:** Same as the Mini Project category. A project exhibition may be conducted.

References:

1. Tom Kelley, *The Art of Innovation: Lessons in Creativity from IDEO, America's Leading Design Firm*, Profile Books, 2002.

Additional References:

1. Tim Brown, *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation*, HarperBusiness, 2009.
2. Jeanne Liedtka, Randy Salzman, and Daisy Azer, *Design Thinking for the Greater Good: Innovation in the Social Sector*, Columbia Business School Publishing, 2017.

Other Useful Design Thinking Frameworks and Methodologies:

1. **Human-Centered Design Toolkit (IDEO)**
<https://www.ideo.com/post/design-kit>
2. **Design Thinking Boot Camp Bootleg (Stanford d.school)**
<https://dschool.stanford.edu/resources/the-bootcamp-bootleg>
3. **Collective Action Toolkit (frog design)**
https://www.frogdesign.com/wp-content/uploads/2016/03/CAT_2.0_English.pdf
4. **Design Thinking for Educators (IDEO)**
<https://designthinkingforeducators.com/>

1901GE254	COMPUTER HARDWARE AND IT ESSENTIALS LAB	L	T	P	C
		0	0	2	1

COURSE OBJECTIVES:

1. To be familiar with Computer Hardware Components and installation of software
2. To learn about searching, downloading, and storing contents in the Cloud Network

COURSE OUTCOMES:

Upon the successful completion of the course, students will be able to

CO1: Perform assembling and disassembling of desktop machine with different peripheral and software installation and servicing

CO2: Simulate data using MS office for Presentation and Visualization.

CO3: Use browsers for searching & accessing/storing the contents to/from cloud.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	2	2	-	-	2
CO2	3	3	2	2	-	-	-	2	2	-	-	2
CO3	3	3	2	2	-	-	-	2	2	-	-	2
CO4	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-
AVG	3	3	2	2	-	-	-	2	2	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	1	-
CO3	1	1	-
CO4	-	-	-
CO5	-	-	-
AVG	1		-

List of Experiment

1. Study of hardware components (such as storage devices, I/O devices, CPU, motherboard, and other peripherals).
2. Installation of operating systems (Windows and Linux).
3. Installation of other software.
4. Study of network components.
5. Network establishment (configuring IP address, Domain Name System).
6. Study of the Internet.
7. Introduction to the web.
8. Usage of internet services — Email, File Sharing, Social Media, etc.
9. Study of firewalls and antivirus software.
10. Troubleshooting various hardware and software problems.

TOTAL:30HOURS

REFERENCES:

1. Craig Zacker & John Rourke, *The Complete Reference: PC Hardware*, Tata McGraw Hill, New Delhi, 2001.
2. Mike Meyers, *Introduction to PC Hardware and Troubleshooting*, Tata McGraw Hill, New Delhi, 2003.
3. B. Govindarajulu, *IBM PC and Clones Hardware Troubleshooting and Maintenance*, Tata McGraw Hill, New Delhi, 2002.
4. R. Kelly Rainer, Casey G. Cegielski, Brad Prince, *Introduction to Information Systems*, 5th Edition, Wiley Publications, 2014.
5. James F. Kurose, *Computer Networking: A Top-Down Approach*, 6th Edition, Pearson, 2012.
6. R. Kelly Rainer, Casey G. Cegielski, Brad Prince, *Introduction to Information Systems*, Fifth Edition, Wiley Publication, 2014
7. Craig Zacker & John Rourke, "The complete reference: PC hardware", Tata McGraw Hill, New Delhi, 2001.

1901GE252	ENGINEERING INTELLIGENCE II	L	T	P	C
		0	0	2	1

Prerequisite: Engineering Intelligence - I

COURSE OBJECTIVES:

To equip students with the knowledge and skills to design, develop, and deploy intelligent systems

Course Outcomes

On successful completion of the course, students will be able to:

CO1: Understand and participate in various vocabulary-building activities.
CO2: Use communication skill workshops effectively for improving reading and writing skills.
CO3: Apply interpersonal skills to motivate creativity and innovation.
CO4: Apply leadership and employability skills to secure career opportunities.
CO5: Prepare a professional resume with all necessary components.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	2	2	-	-	-	2	2
CO2	3	3	2	2	-	2	2	-	-	-	2	2
CO3	3	3	2	2	-	2	2	-	-	-	2	2
CO4	3	3	2	2	-	2	2	-	-	-	2	2
CO5	3	3	2	2	-	2	2	-	-	-	2	2
AVG	3	3	2	2	-	2	2	-	-	-	2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	2	-
CO2	-	2	-
CO3	-	2	-
CO4	-	2	-
CO5	-	2	-
AVG	-	2	-

MODULE I	VOCABULARY BULIDING	6 Hours
Parts of Grammar-SVA-Art of Writing-word building activities		
MODULE II	COMMUNICATION WORKSHOP	6 Hours
Story Telling – News paper Reading - Extempore.		
MODULE III	INTERPERSONAL SKILLS	6 Hours
Personality Development - Creativity and innovation – Critical Thinking and Problem Solving – Work Ethics-Technical Skill vs Interpersonal Skills		
MODULE IV	LEADERSHIP & EMPLOYABILITY SKILLS	6 Hours
Levels of Leadership -Making of leader-Types of leadership – Transactions Vs Transformational Leadership – Exercises – Industry Expectations & Career Opportunities – Recruitment patterns.		

MODULE V	RESUME BUILDING	6 Hours
Importance of Resume – Resume Preparation – introducing on self		
TOTAL:30 HOURS		
REFERENCES:		
1. Barun K. Mitra, <i>Personality Development & Soft Skills</i> , First Edition, Oxford Publishers, 2011.		
2. Raymond Murphy, <i>Essential English Grammar in Use</i> , Cambridge University Press, New Delhi, Third Edition, 2007.		
3. Arun Sharma and Meenakshi Upadhyay, <i>How to Prepare for Verbal Ability and Reading Comprehension for CAT</i> , McGraw Hill Publication, Seventh Edition, 2017.		

1901GEX51	CAD (COMPUTER AIDED DRAFTING) LAB	L	T	P	C
		0	0	2	1

Prerequisite:

1. Basic Computer knowledge
2. Engineering Graphics

COURSE OBJECTIVES:

1. To develop in students, Drafting skills for communication of concepts, ideas and design of Engineering products
2. To expose them to existing national standards related to technical drawings

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1;Ability to use the software packers for drafting and modeling

CO2: Learned basic concept to drawing, edit, dimension, hatching etc. to develop 2& 3D Modelling.

CO3:Able to create front view and top view of simple solids

CO4: Able to create isometric projection of simple objects.

CO5:Able to Create 3D models of Simple Objects and obtaining 2-D multi-view drawings from 3-D model

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	2	-	-	-	2	-	2	1
CO2	3	3	2	-	2	-	-	-	2	-	2	1
CO3	3	3	2	-	2	-	-	-	2	-	2	1
CO4	3	3	2	-	2	-	-	-	2	-	2	1
CO5	3	3	2	-	2	-	-	-	2	-	2	1
AVG	3	3	2	-	2	-	-	-	2	-	2	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	2	2
CO2	-	2	2
CO3	-	2	2
CO4	-	2	2
CO5	-	2	2
AVG	-	2	2

Basics commands of a CAD software-two-dimensional drawing, editing, layering and dimensioning - coordinate Systems-Drawing practice - orthographic views of simple solids using CAD software.

1. Study of software capabilities for drafting and modeling — coordinate systems (absolute, relative, polar, etc.) — creation of simple figures like polygons and general multi-line figures.
2. Drawing a title block with necessary text and projection symbols.
3. Drawing curves such as parabolas, spirals, involutes using B-spline or cubic spline methods.
4. Drawing front view and top view of simple solids like prism, pyramid, cylinder, cone, etc., with proper dimensioning.
5. Drawing front view, top view, and side view of objects from given pictorial views (e.g., V-block, base of a mixer, simple stool, objects with holes and curves).
6. Drawing sectional views of prisms, pyramids, cylinders, cones, etc.
7. Drawing isometric projections of simple objects.

8. Creating 3-D models of simple objects and generating 2-D multi-view drawings from the 3-D models.

Total: 30 Hours

REFERENCES:

1. N. D. Bhatt, Machine Drawing, Charotar Publishing House Pvt. Ltd., 2014.
2. P. S. Gill, A Textbook of Machine Drawing, Katson Books, 2013.
3. R. K. Dhawan, A Textbook of Machine Drawing, S. Chand, 2012.
4. K. C. John, Textbook of Machine Drawing, PHI Learning Pvt. Ltd., 2009.

1901GEX01	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY (Common for all UG programmes)	L	T	P	C
		0	0	2	1

COURSE OBJECTIVES: To apply the fundamentals of Electrical and Electronics Engineering

PREREQUISITES:

1. Verification of Ohm's Law and Kirchhoff's Laws.
2. Experiments involving logic gates.
3. Fan and light control using regulators.
4. Design of a 6V regulated power supply.
5. Energy conservation demonstration using an energy meter.
6. Waveform generation and calculation of RMS and average values.
7. Experiments based on IC 555 and IC 741.
8. Experiments on earthing.
9. Staircase wiring and residential building wiring.
10. Speed control of DC shunt motor.

COURSE OUTCOMES

Upon completion of this course, students will be able to:

CO1: Design and analyze electronic circuits.

CO2: Test digital logic gates.

CO3: Control lights and speed of motors.

CO4: Measure electrical parameters using instruments.

CO5: Generate waveforms and construct different wiring schemes.

COs Vs POs MAPPING:22

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	2	-	-	2	2	-	-	2
CO2	3	3	2	-	2	-	-	2	2	-	-	2
CO3	3	3	2	-	2	-	-	2	2	-	-	2
CO4	3	3	2	-	2	-	-	2	2	-	-	2
CO5	3	3	2	-	2	-	-	2	2	-	-	2
AVG	3	3	2	-	2	-	-	2	2	-	-	2

COs Vs PSO MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	-	-
CO2	1	-	-
CO3	1	-	-
CO4	1	-	-
CO5	1	-	-
AVG	1	-	-

REFERENCES (BOOKS):

1. Edward Hughes, *Electrical Technology*, Pearson Education.
2. D. P. Kothari and Nagrath, *Basic Electronics*, McGraw Hill Education, 2013.
3. Paul Scherz and Simon Monk, *Practical Electronics for Inventors*, McGraw Hill Publications, 2013.

REFERENCES (WEBSITES):

1. <https://nptel.ac.in/courses/122106025/>

1901PHX51	ENGINEERING PHYSIC SLAB				L	T	P	C				
					0	0	2	1				
PREREQUISITE: Basic knowledge in physics												
COURSE OBJECTIVES:												
1. To learn the proper use of various kinds of physics laboratory equipment 2. To learn how data can be collected, presented and interpreted in a clear and concise manner. 3. To determine error in experimental measurements and techniques used to minimize such error. 4. To make the student an active participant in each part of all lab exercises												
COURSE OUTCOMES:												
At the end of thiscourse, Students will be able to												
1. Utilize the concept of twisting couple to find the Rigidity Modulus and Moment of Inertia of a wire 2. Experiment with properties of materials to find the Young’s modulus of the material under uniform bending 3. Choose the concept of streamline flow of liquids in capillary tubes and measure the viscosity of liquids 4. Test the phenomenon of interference of light by forming fringes and find the thickness through air-wedge method. 5. Determine the particle size and wavelength of laser source through diffraction phenomenon												
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	2	2	-	-	-
CO2	3	3	-	2	-	-	-	2	2	-	-	-
CO3	3	3	-	2	-	-	-	2	2	-	-	-
CO4	3	3	-	2	-	-	-	2	2	-	-	-
CO5	3	3	-	2	-	-	-	2	2	-	-	-
AVG	3	3	-	2	-	-	-	2	2	-	-	-
COs Vs PSOs MAPPING:												
COs	PSO1	PSO2	PSO3									
CO1	2	-	-									
CO2	2	-	-									
CO3	2	-	-									
CO4	2	-	-									
CO5	2	-	-									
AVG	2	-	-									

List of Experiments:

1. Determination of wavelength of various colours of mercury spectrum using laser grating.
2. Determination of velocity of liquids using ultrasonic interferometer.
3. Determination of the dispersive power of a prism using spectrometer.
4. Determination of the unknown resistance of the given wire using Carey-Foster's Bridge.
5. Determination of the band gap of the given semiconductor.
6. Determination of the acceptance angle and particle size using laser.
7. Torsional pendulum – Measurement of rigidity modulus of a steel wire.
8. Measurement of thickness of a thin wire using Air Wedge method.
9. Measurement of Young's modulus – Uniform and non-uniform bending.
10. Measurement of thermal conductivity using Lee's Disc method.

References:

1. Practical Physics", R.K. Shukla, AnchalSrivastava, New age international (2011)
2. B.Sc. Practical Physics", C.L Arora, S. Chand &Co. (2012)